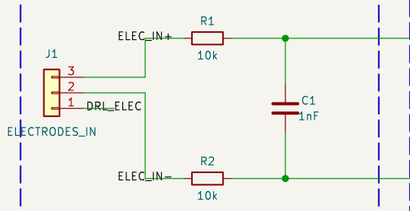
1. INPUT & RF FILTER

Diff & CM RF Filtering

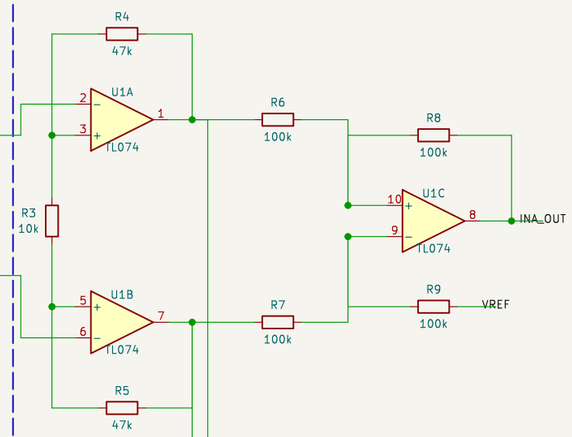


Patient Current Limit: 10k

Differential RF Cutoff: ~8 kHz

2. STAGE 1: INA PRE-AMP (11x)

TL074 JFET Buffers + Difference Amp

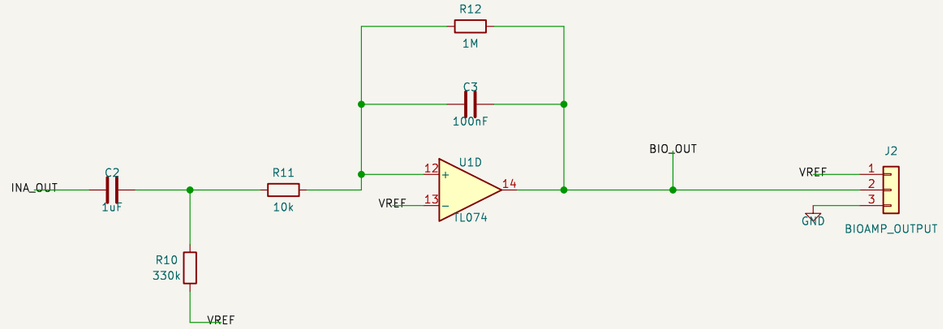


Gain Av1 = 1 + 2*(47k)/10k = 10.4x

Matched 100k Bridge CMRR > 90 dB

3. STAGE 2: ACTIVE FILTER & OUTPUT

0.48Hz HPF + 100x Post-Amp + 40Hz LPF (Total: 1040x)

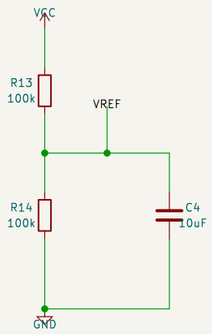


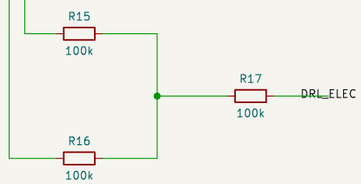
DC-Blocking HPF: fc = 1 / (2*pi*330k*1uF) = 0.48 Hz

Active LPF: fc = 1 / (2*pi*1M*100nF) = 40 Hz (Av2 = 100x)

4. VREF (2.5V) & ACTIVE DRL REF GENERATOR

Low-Noise VREF Bias + Inverted Common-Mode Feedback





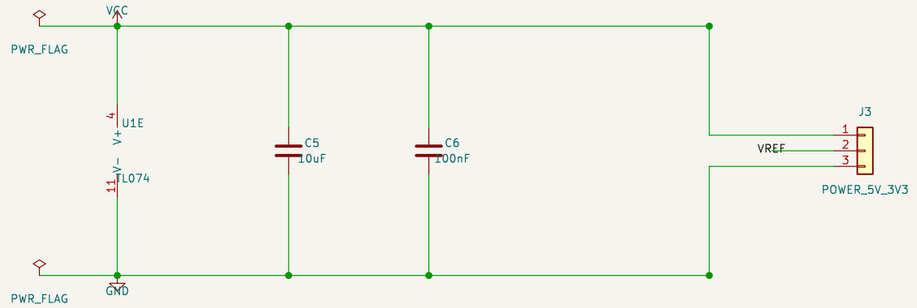
Low-Noise VREF = VCC / 2 (2.5V @ 5V)

DRL Common-Mode Cancellation

Patient Safe Current Limit: R17 = 100k

5. POWER MANAGEMENT & TL074 RAILS

Bulk 10uF Tantalum + 100nF Ceramic HF Bypass



Single Supply: +3.3V to +5.0V

Bulk 10uF Tantalum + 100nF High-Frequency Bypass

CERN-OHL Open Hardware Architecture
Standard KiCad 8 Native Professional Schematic
Fully Annotated & PCB Ready KiCad 8 Architecture
Full 1040x Gain ECG / EMG / EEG AFE with Active DRL & ESD Clamping
Universita degli Studi di Palermo – Bioelectronics Lab

Sheet: /
File: UniPa_BioAmp_Pro.kicad_sch

Title: UniPa BioAmp Pro – Biomedical Biopotential Analog Front-End

Size: A3	Date: 2026-09-26	**Rev: v1.1**
KiCad E.D.A. 8.0.8		Id: 1/1